

Observational Paper #1: Cassini RSS & VIMS Radio/Optical Occultation Analysis of Saturn’s Ring Resonances and Shepherd Moon Dynamics

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Abstract

We perform an observational analysis of Saturn’s ring structure using high-precision radio (RSS) and optical (VIMS) occultation datasets from the Cassini spacecraft. We apply first-principles Lindblad resonance theory (Goldreich & Tremaine 1978, 1979) including Saturn’s zonal oblateness ($J_2 = 0.01629$) to model the sharp ring boundaries. We find that the inner edge of the Cassini Division ($r_{\text{obs}} = 117,580$ km) matches the Mimas 2:1 Inner Lindblad Resonance (ILR) ($r_{\text{calc}} = 117,186$ km, RSE = 0.3%), while the sharp outer edge of the main A-Ring ($r_{\text{obs}} = 136,770$ km) is dynamically maintained by the Janus 7:6 ILR ($r_{\text{calc}} = 136,928$ km, RSE = 0.1%). Furthermore, modeling the net resonant shepherd torques exerted by Prometheus and Pandora on the narrow F-Ring core yields a torque balance location of $r_{\text{calc}} = 140,572$ km, reproducing Cassini RSS observations ($r_{\text{obs}} = 140,220$ km, RSE = 0.25%).

1 Introduction & Physical Formulation

Saturn’s main rings are truncated and structured by gravitational resonances with embedded and external satellites. An Inner Lindblad Resonance (ILR) occurs at ring radius r_{res} where orbital frequency $\Omega(r)$ and epicyclic frequency $\kappa(r)$ satisfy:

$$m\Omega(r) - \kappa(r) = m_{\text{moon}}n_{\text{moon}} \quad (1)$$

Including Saturn’s quadrupolar oblateness J_2 :

$$r_{\text{ILR}} \approx a_{\text{moon}} \left(\frac{m_{\text{moon}}}{m_{\text{ring}}} \right)^{2/3} \left[1 + 0.6J_2 \left(\frac{R_S}{r_{\text{kepler}}} \right)^2 \right] \quad (2)$$

For shepherd satellites Prometheus (inner) and Pandora (outer) confining the F-Ring, the net resonant torque balance condition $\dot{L}_{\text{net}} = T_{\text{Prometheus}} + T_{\text{Pandora}} = 0$ yields:

$$r_{\text{core}} = \frac{a_{\text{Prom}} + (M_{\text{Prom}}/M_{\text{Pan}})^{1/4} a_{\text{Pan}}}{1 + (M_{\text{Prom}}/M_{\text{Pan}})^{1/4}} \quad (3)$$

2 Observational Results & Model Comparison

Using our C++ engine target `//:saturn_ring_resonances_paper`, we compare theoretical predictions against Cassini occultation measurements:

The close agreement ($R^2 > 0.999$) confirms that satellite Lindblad torques and shepherd mass ratios govern Saturn’s major ring boundaries.

Table 1: Cassini Occultation Measurements vs First-Principles Resonant Model

Ring Feature	Responsible Satellite / Resonance	Observed r (km)	Model r (km)	Relative Error
Cassini Division Edge	Mimas 2:1 ILR	117,580	117,186	0.33%
A-Ring Outer Edge	Janus 7:6 ILR	136,770	136,928	0.11%
F-Ring Core	Prometheus / Pandora Shepherds	140,220	140,572	0.25%